

AegisWeb: Final Project Outline & Architecture Report

1. Executive Summary

Project Name: AegisWeb

Objective: To orchestrate a full-stack, cybersecurity-first web application that bridges premium user-interface design with rigorous backend authentication, dynamic session management, and automated intrusion detection reporting.

2. Technology Stack & Foundation

Framework: Next.js (App Router) — Providing seamless hybrid routing and secure backend API handling.

Database Architecture: SQLite integrated with the Prisma ORM (v5 LTS) for stable, deterministic, and type-safe data migrations.

Visual Design: A custom-built, vanilla CSS Glassmorphism aesthetic eliminating heavy CSS-framework abstraction layers.

Telemetry Charts: Recharts — Powering the dashboard's data plotting.

3. Core Architecture & Features Developed

3.1 Advanced Authentication Pipeline

Salting & Hashing: Uncompromising implementation of bcryptjs hashing. Plaintext passwords never touch the database.

JWT Session Protocol: Stateless, rapidly verifiable JSON Web Tokens served exclusively through HttpOnly, production-secure cookies thwarting XSS theft vectors.

Password Strength Checker: A dynamic interface actively interrogates the user's password input alongside a strict backend Regex wall enforcing 8+ characters, symbols, and numbers.

3.2 Real-Time Intrusion Detection System (IDS)

Aggressive IP Telemetry: Every single interaction on the /api/auth/login endpoint parses and tracks proxy-forwarded IP headers seamlessly.

Account Lockout System: A rapid 15-minute early-exit lockout mechanism blocks highly targeted Brute-Force and Dictionary attacks. It cuts off connections if an account is pounded with 5 failing attempts, preventing heavy DB queries entirely.

Suspicious Activity Triaging: Volumetric anomalies spawn immediate telemetry warnings written to the database labeled systematically as HIGH or CRITICAL severity incidents depending on whether it is IP-based or User-based targeting.

3.3 Security Analytics Dashboard

Telemetry Visualizations: Interactive Recharts display a precise 7-day trailing lookback aggregating Successful Authentications vs Threat Rejections.

Last Login Verification: Securely extracts username payloads from JWTs to cross-reference and permanently display the user's previous successful entry-point along with the historical IP address to quickly recognize session hijacking.

CSV Data Exfiltration: A direct, single-click endpoint that collates the entirety of the incident tables into downloadable, structurally compliant Comma-Separated Values (CSV) formats for raw spreadsheet / SIEM pipeline ingestion.

4. Development Highlights

Robustness Against Dependency Drift: Flawlessly circumnavigated major breaking-version changes inside the Prisma 7 client, restricting the environment locally to LTS tooling to ensure consistent compilation.

Type-Safe Ecosystem: Handled and mitigated implicit TypeScript compilation errors (any typing in analytics logic) resulting in a pristine next build validation output.

5. Conclusion

AegisWeb acts as an optimal, modern technical exhibition highlighting how to establish zero-trust, resilient, and continuously monitored infrastructures strictly within the bleeding-edge Next.js environment.